

Toward a Sparse Interpretable Audio Codec

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Abstract—Most widely-used modern audio codecs, such as Ogg Vorbis and MP3, as well as more recent “neural” codecs like Meta’s Encodec or Descript’s are based on block-coding; audio is divided into overlapping, fixed-size “frames” which are then compressed. While they result in excellent reproduction quality and can be used for downstream tasks such as text-to-audio, they do not produce an intuitive, directly-interpretable representation. In this work, we introduce a proof-of-concept audio encoder that represents audio as a sparse set of events and their times-of-occurrence. Rudimentary physics-based assumptions are used to model attack and the physical resonance of both the instrument being played and the room in which a performance occurs, hopefully encouraging a sparse, parsimonious, and easy-to-interpret representation.

1. INTRODUCTION

We imagine a near-future audio codec where some types of musical composition could take place directly in the audio codec space. Text-to-audio is an excellent interface for non-musicians creating background music for movies, advertisements or social media content, but it is our view that experienced musicians and composers work in a “space” not fully captured by language and will prefer a much finer-grained representation that affords an effectively infinite range of possible sounds.

This work clearly does not replace current music generation models, but could serve as the underlying encoding on which generative models are trained. It is the authors’ intuition that models trained on this rich, symbolic representation might have a much deeper “understanding” of the content they produce, given the point-cloud-like nature of the signal. Instead of predicting the next frame, the generative model would be predicting the relationships between “events”. Long-term coherence in musical generation has always been a difficult problem, and we speculate that many models spend enormous shares of their capacity learning to reproduce physical resonance, rather than the human forces that drive them in interesting, musical directions.

All model and experiment code is written in PyTorch and can be found on GitHub¹. Audio examples can be heard at this [https URL](https://github.com/JohnVinyard/matching-pursuit)²

2. PRIOR WORK

This work draws inspiration from a number of existing techniques, including the field of Auditory Scene Analysis and blind source separation, as well as machine learning techniques like matching pursuit and synthesis techniques like granular synthesis.

- auditory scene analysis - blind source separation - audio source separation - granular synthesis - matching pursuit - source-excitation models (NEEDS A REFERENCE)

3. DATA SET

We train our model on the MusicNet dataset [1], an open dataset containing 33 hours of classical music, recorded in diverse spaces using a range of recording equipment of varying quality. The recordings often include leading silence, trailing applause, and incidental human sounds throughout (coughs, movement, etc).

¹<https://github.com/JohnVinyard/matching-pursuit>

²<https://blog.cochlea.xyz/sparse-interpretable-audio-codec-paper.html>

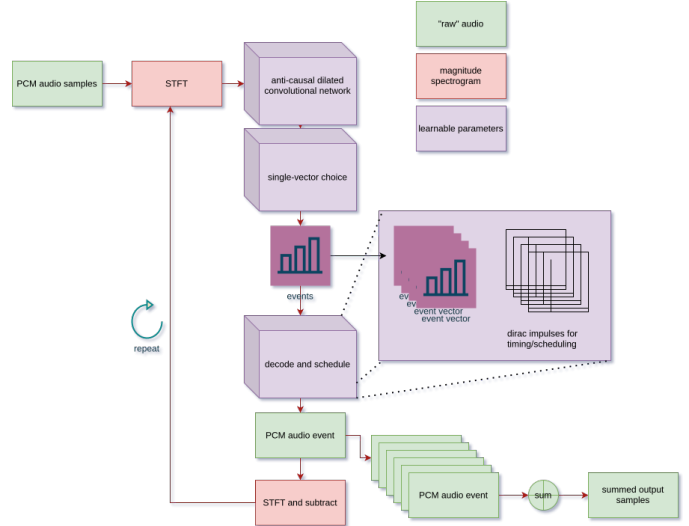


Fig. 1: Model architecture

4. MODEL

4.1. Encoder

TODO: Consider a U-NET or Transformer encoder paired with a sparsity loss

Inspired by matching pursuit, the encoding process is iterative, or incremental. At each step, the encoder chooses a single event vector and time-of-occurrence and passes this to the decoder, which generates and schedules audio and subtracts it from the input representation. This process is repeated for some fixed number of steps or until some stopping condition is reached.

The encoder is parameterized as an anti-causal, dilated convolutional network.

4.2. Decoder

In order to encourage a sparser and more easy-to-interpret representation, we parameterize the decoder as a neural network with the following structure:

Each event is roughly interpreted as a physical process, including the injection of energy into some system and the long convolutions representing the resonance of the system and the way the system is deformed over time as it resonates.

TODO: Diagram of decoder block and scheduling process

5. EXPERIMENTS

5.1. Training Algorithm

During training, we repeatedly draw segments of audio from the Music Net dataset at random, with length 2^{17} samples at 22050hz sampling rate and a batch size of two. Each batch is passed through the encoding and decoding process for a fixed number of steps, 32, in our case, and then an iterative loss is computed, encouraging each

step to have removed as much energy (expressed as the L1 norm) from the signal as possible. This loss is minimized using the Adam optimizer with a learning rate of $1e^{-4}$.

5.2. Loss Functions

6. INTERPRETABLE AND MANIPULATABLE REPRESENTATION

7. FUTURE WORK

8. DOUBLE-BLIND REVIEW

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To achieve the best rendering in the proceedings, we strongly encourage you to use Times Roman font. In addition, this will give the proceedings a more uniform look. Use a font that is no smaller than 9pt type (the recommended setting) for the body text. If you wish to use 10pt type font, remove the 9pt option when loading the document class. Note that the abstract, figure captions, and tables are in 8pt font in both cases.

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Major headings, for example, "**1. INTRODUCTION**", should appear in all capital letters, bold face, centered in the column. Vertical spacing before the heading should be between 0.055 inches and 0.166 inches in 9pt font, and between 0.062 inches and 0.187 inches in 10pt font; vertical spacing after the heading should be between 0.039 inches and 0.094 inches in 9pt font, and between 0.044 inches and 0.106 inches in 10pt font. Use a period (".") after the heading number. Sections can be referenced using [Section 16](#), [Section 16.1](#), and [Section 16.1.1](#).

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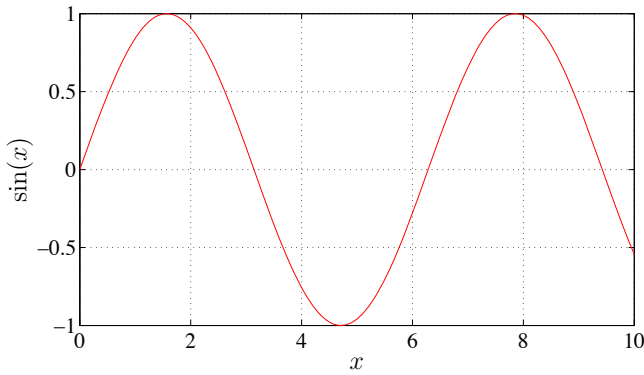


Fig. 2: Example of a figure with experimental results, where the caption is long and wraps on the next line.

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When inserting PDFs, we recommend using `pdfcrop` to remove unnecessary margins. This will allow you to better fine-tune the size of your figure when inserting it with `\includegraphics`.

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Equations should be placed on separate lines and consecutively numbered with equation numbers in parentheses flush with the right margin, as illustrated in (1) that gives the homogeneous acoustic wave equation in Cartesian coordinates [3],

$$\Delta^2 p(\mathbf{x}, t) - \frac{1}{c^2} \frac{\partial^2 p(\mathbf{x}, t)}{\partial t^2} = 0, \quad (1)$$

where $p(\mathbf{x}, t)$ is an infinitesimal variation of acoustic pressure from its equilibrium value at position $\mathbf{x} \in \mathbb{R}^3$ and time t , and where c denotes the speed of sound.

Symbols in your equation should be defined before the equation appears or immediately following. Use (1), not Eq. (1) or equation (1), except at the beginning of a sentence: “Equation (1) is ...”

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³or not at all!

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REFERENCES

- [1] J. Thickstun, Z. Harchaoui, and S. M. Kakade, “Learning features of music from scratch,” in *International Conference on Learning Representations (ICLR)*, 2017.
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- [3] E. Williams, *Fourier Acoustics: Sound Radiation and Nearfield Acoustic Holography*. London, UK: Academic Press, 1999.
- [4] C. Jones, A. Smith, and E. Roberts, “A sample paper in conference proceedings,” in *Proc. ICASSP*, vol. II, Apr. 2003, pp. 803–806.
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